

A Machine-Certified Closure of Erdős Problem #1212

Public minimal proof, 14 July 2026

Abstract

We prove the original statement of Erdős Problem #1212: the positive visible lattice, after deletion of every prime–prime vertex, contains an unbounded unit-step path. The proof first reduces the path to tightness of bad separating contours. Fixed-contour prime–prime contributions have zero translation density, so compatible prime-divisibility witnesses give a positive, coefficientwise necklace expansion. Splitting that expansion at the first repeated prime gives an all-fresh term and a repeat operator. The all-fresh term is summable. A certified tilted Hashimoto inequality bounds all repeated primes at least 61 by 0.067. For the twelve remaining primes, exact Fourier projection, exact-rational root enclosures, binary64 centres, directed-rounding tails, and a full-state interval comparison prove the common-cone bound

$$\mathcal{R}_{<61}u \leq U \leq 0.91717912709094451u < 0.933u.$$

The comparison covers all 55,076,704 allocated states and has zero threshold, positive-over-zero, negative-upper, and nonfinite failures. Thus the complete first-repeat norm is strictly below one, separating contour mass cannot escape, and the required path exists. This note contains only that closed proof chain and its machine-certificate contract.

1 The unchanged target

Let

$$\mathcal{G} = \{(x, y) \in \mathbf{N}^2 : \gcd(x, y) = 1\}$$

with an edge when exactly one coordinate changes by 1 or -1 , and put

$$\mathcal{S} = \{(x, y) \in \mathcal{G} : x > 1, y > 1, \text{ and at least one of } x, y \text{ is composite}\}.$$

Thus $\mathcal{S}$ is obtained from the positive visible lattice by deleting the two coordinate boundary strips and every prime–prime vertex.

Theorem 1.1 (Erdős #1212). *There is a sequence $(P_t)_{t \geq 0}$ in $\mathcal{S}$ such that consecutive terms are adjacent and, for every $R \geq 1$, some P_t has a coordinate at least R .*

Proposition 1.2 (Finite reachability dictionary). *Theorem 1.1 is equivalent to the existence of one $v \in \mathcal{S}$ from which finite paths in $\mathcal{S}$ leave every finite box.*

Proof. An infinite path gives the finite paths by truncation. Conversely, remove loops from each finite path. The rooted tree of finite simple paths beginning at v is locally finite and has arbitrarily large depth. König’s infinity lemma gives an infinite ray, which cannot remain in a finite box. $\square$

2 Fixed contours and the only escape event

Put

$$\mathcal{D} = \{(x, y) \in \mathbf{N}^2 : \gcd(x, y) > 1\}, \quad \mathcal{P}^{(2)} = \{(p, q) : p, q \text{ are prime}\}.$$

Away from the boundary strips, $\mathbf{N}^2 \setminus \mathcal{S} = \mathcal{D} \cup \mathcal{P}^{(2)}$.

Let $A \subset \mathbf{Z}^2$ be finite. A map from A to the primes is *compatible* when equal labels p occur only at sites congruent coordinatewise modulo p . Give such a map the weight

$$\omega(w) = \prod_{p \in w(A)} p^{-2}, \quad Z(A) = \sum_{w \text{ compatible}} \omega(w).$$

The sum is finite: a fibre of size at least two can be labelled only by a prime dividing both coordinate differences of two sites, while singleton fibres contribute the convergent sum $\sum_p p^{-2}$.

Proposition 2.1 (Exact witness expansion). *For*

$$H_A(T) = \{z \in [1, T]^2 : z + a \in \mathcal{D} \text{ for every } a \in A\},$$

one has

$$\limsup_{T \rightarrow \infty} \frac{|H_A(T)|}{T^2} \leq Z(A).$$

A fixed compatible witness w is realized with natural density exactly $\omega(w)$.

Proof. Choose one common prime divisor at every translated site. Equal chosen primes force compatibility. Conversely, compatibility and the Chinese remainder theorem give one residue class in each coordinate modulo $\prod_{p \in w(A)} p$, of density $\omega(w)$. Restricting first to primes at most R , applying the union bound, and then sending R to infinity justifies the infinite prime sum. $\square$

Theorem 2.2 (Fixed-contour prime cancellation). *Let $B_A(T)$ count translations $z \in [1, T]^2$ for which every $z + a$ is outside $\mathcal{S}$. Boundary translations contribute only $O_A(T)$, and*

$$\frac{|B_A(T)| - |H_A(T)|}{T^2} \rightarrow 0.$$

Thus the translation density of a fixed bad contour is exactly the density of the same contour made entirely of nonvisible sites.

Proof. The inclusion $H_A(T) \subseteq B_A(T)$ is immediate. A translation in the difference has at least one prime–prime site. For fixed a there are $O_A(T^2 / \log^2 T)$ such translations, and a union bound over A proves the claim. $\square$

Let C_∞ be Vardi's unique infinite component of the full visible lattice; it has a positive asymptotic density θ [2]. Let $\mathfrak{C}$ be the simple closed star-contours surrounding the origin. For $M \geq 1$ let $E_{T,M}$ be the set of $z \in C_\infty \cap [1, T]^2 \cap \mathcal{S}$ for which some $\gamma \in \mathfrak{C}$ with $|\gamma| > M$ satisfies $z + \gamma \subseteq \mathbf{N}^2 \setminus \mathcal{S}$, and put

$$\tau(M) = \limsup_{T \rightarrow \infty} \frac{|E_{T,M}|}{T^2}.$$

Theorem 2.3 (No escape closes the problem). *If $\tau(M) \rightarrow 0$ as $M \rightarrow \infty$, then Theorem 1.1 holds.*

Proof. Assume every component of $\mathcal{S}$ is finite. The planar outer-boundary lemma gives a simple closed bad star-contour around every $z \in \mathcal{S}$. For fixed M , there are finitely many relative contour shapes of length at most M . A short contour around a point of C_∞ cannot consist entirely of sites in $\mathcal{D}$, since an infinite visible path from that point must cross it. It therefore contains a prime–prime site. By Theorem 2.2, the roots admitting any of the finitely many short shapes have density zero. Prime–prime roots also have density zero, while $C_\infty \cap \mathcal{S}$ has density $\theta > 0$. Hence $\theta \leq \tau(M)$ for every M , a contradiction. Thus $\mathcal{S}$ has an unbounded component, and Proposition 1.2 extracts the path. $\square$

3 Positive first-repeat renewal

Fix $Y = 11$, put $Q = 2310$, and let

$$F_{11} = \{(a, b) : \gcd(a, b, Q) > 1\}.$$

Contract only the finite star-components of F_{11} and retain every site outside F_{11} , together with all component–site incidences and external star-edges. On directed incidences use the Hashimoto update that forbids the immediate reverse edge and charges a factor q upon entering an external site. Write

$$H(q) = A + qB,$$

where B consists of transitions entering an external site.

For distinct tail primes define

$$d_m = \sum_{\substack{p_1, \dots, p_m > 11 \\ p_i \text{ distinct}}} \frac{1}{p_1^2 \cdots p_m^2}.$$

Lemma 3.1 (All-fresh summability). *For every $C > 0$, $\sum_{m \geq 0} C^m d_m < \infty$.*

Proof. Let $e_m = d_m/m!$. For $t > 0$,

$$\sum_{m \geq 0} e_m t^m \leq \prod_{n \geq 1} \left(1 + \frac{t}{n^2}\right) = \frac{\sinh(\pi\sqrt{t})}{\pi\sqrt{t}} \leq e^{\pi\sqrt{t}}.$$

Thus $e_m \leq e^{\pi\sqrt{t}} t^{-m}$. Taking $\sqrt{t} = 2m/\pi$ and using $m! \leq m^m$ gives

$$d_m \leq \left(\frac{e^2 \pi^2}{4m}\right)^m,$$

which is superexponentially summable against C^m . $\square$

Read the tail-prime labels in order and stop at the first repeated label. Let $\mathcal{F}$ be the all-fresh Green function and $\mathcal{R}$ the positive operator of compatible first-repeat excursions, retaining the lifted displacement. The complete necklace Green function satisfies the coefficientwise renewal inequality

$$\mathcal{G} \preceq \mathcal{F} + \mathcal{R}\mathcal{G}. \tag{1}$$

No cancellation is used.

Lemma 3.2 (Subcritical renewal implies no escape). *If $\mathcal{F}$ is finite in the rooted area-weighted necklace norm and $\|\mathcal{R}\| < 1$ in the same positive norm, then $\tau(M) \rightarrow 0$.*

Proof. Iterating (1) coefficientwise gives

$$\mathcal{G} \preceq \sum_{j \geq 0} \mathcal{R}^j \mathcal{F}, \quad \|\mathcal{G}\| \leq \frac{\|\mathcal{F}\|}{1 - \|\mathcal{R}\|} < \infty.$$

Every compatible translated bad contour contributes to the nonnegative witness expansion bounded by this necklace series. The contribution of contours of length greater than M is the tail of a finite nonnegative series, hence tends to zero. Proposition 2.1 turns the same bound into translation upper density, while Theorem 2.2 removes fixed-shape prime–prime terms. Therefore the defining upper density $\tau(M)$ tends to zero. $\square$

4 The infinite repeat-prime tail

Put

$$q_{\text{tr}} = \sum_{p > 11} \frac{1}{p^2}, \quad \bar{q} = \frac{22751}{10^6}.$$

Lemma 4.1 (Translation-tail majorant). *One has $q_{\text{tr}} < \bar{q}$.*

Proof. Every prime greater than 11 is coprime to 2310, and the only integer in $[1, 11]$ coprime to 2310 is 1. Therefore

$$q_{\text{tr}} \leq \frac{\pi^2}{6} \prod_{p \leq 11} \left(1 - \frac{1}{p^2}\right) - 1 \leq \frac{1722359}{75707401} < \frac{22751}{10^6},$$

where the middle inequality uses $\pi < 355/113$. $\square$

For a lifted displacement d and $\xi \in \mathbb{R}^2$, let $H_\xi(q)$ multiply the transition by $e^{\langle \xi, d \rangle}$.

Theorem 4.2 (Certified translation kernel). *On the cyclic support of the 2310-period incidence graph,*

$$\rho(H(\bar{q})) < 0.97, \quad \rho(H_\xi(\bar{q})) < 0.97$$

for $\xi = (\pm 3/20, 0), (0, \pm 3/20)$.

Proof. The finite graph has 333,280 permanent-component types, 3,317,760 external sites, 12,387,904 directed component–site states, and 14,407,568 directed external-site states. The largest permanent component has 77 sites and coordinate span 12.

Power iteration only constructs a positive witness v . Acceptance is the statewise Collatz inequality

$$H_\xi(\bar{q})v \leq 0.97v.$$

Every binary64 source sum is recomputed in long-double arithmetic, enlarged by a relative 10^{-10} , and compared with a target reduced by the same amount. Tilt factors are rational upper enclosures obtained from Taylor intervals. The untilted maximum upper ratio is 0.963345 and the $(3/20, 0)$ ratio is 0.965210, with zero failures. Reflections and coordinate interchange give the remaining tilts. The source hash and exact transcript are frozen in `certificate/hashimoto.txt`; the checker is `kernel/skeleton_winding_kernel.cpp`. $\square$

Let $\mathcal{R}_{\geq 61}$ be the first-repeat contribution whose repeated label is a prime at least 61.

Theorem 4.3 (Large repeat primes). *In the certificate norm of Theorem 4.2,*

$$\|\mathcal{R}_{\geq 61}\| < 0.067.$$

Proof. Two sites carrying the same prime p differ by a nonzero vector in $p\mathbf{Z}^2$. One of the four oriented tilts contributes at most $e^{-3p/20}$. Removing both scalar endpoint charges conservatively gives

$$\|\mathcal{R}_{\geq 61}\| \leq \frac{4}{\bar{q}^2(1-0.97)} \sum_{p \geq 61} \frac{e^{-3p/20}}{p^2}.$$

Replacing primes by all integers and summing geometrically,

$$\sum_{p \geq 61} \frac{e^{-3p/20}}{p^2} \leq \frac{e^{-183/20}}{61^2(1-e^{-3/20})} < \frac{23}{24000 \cdot 61^2}.$$

Consequently

$$\|\mathcal{R}_{\geq 61}\| < \frac{1150000000000}{17334174345489} < 0.067.$$

□

5 The finite twelve-prime certificate

Only

$$\{13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59\}$$

remain. Decompose them as

$$S_0 = \{13, 17, 19, 23, 29\}, \quad S_1 = \{31, 37, 41, 43, 47, 53, 59\}, \quad s_i = \sum_{p \in S_i} p^{-2}.$$

For a character χ of $(\mathbf{Z}/p\mathbf{Z})^2$, twist every lifted displacement and write the operators as $H_\chi(\bar{q})$ and B_χ . Character orthogonality gives the exact projector

$$\frac{1}{p^2} \sum_{\chi} \chi(d) = \mathbf{1}_{d \in p\mathbf{Z}^2}.$$

Together with the common-prime cost, the coefficient is p^{-4} .

Let $\eta_m = d_m/\bar{q}^m$. It is decreasing, because $d_{m+1} \leq \bar{q}d_m$. For nonnegative coefficients c_n define

$$\mathcal{Q}_{S,N}(c) = \sum_{p \in S} \frac{1}{p^4} \sum_{\chi} B_\chi \left(\sum_{n=0}^N c_n H_\chi(\bar{q})^n \right) B,$$

where $c_n = 1$ for $n \leq 20$ and $c_n = \eta_{\lfloor n/2 \rfloor}$ thereafter.

The common-cone vector u is strictly positive on every used state. Let C_i be the binary64 centre of $\mathcal{Q}_{S_i, N_i}(c)u$, with $N_0 = 40$ and $N_1 = 24$, and define the directed-rounding positive images

$$\begin{aligned} A_i &= B \left(\sum_{n=0}^{N_i} c_n H(\bar{q})^n \right) Bu, \\ T_{25} &= B \left(\sum_{n=25}^{400} \eta_{\lfloor n/2 \rfloor} H(\bar{q})^n \right) Bu, \\ T_{41} &= B \left(\sum_{n=41}^{400} H(\bar{q})^n \right) Bu. \end{aligned}$$

All 16,548 characters are included. The root tables are verified without floating trigonometry: Machin's identity and rational alternating arctangent series enclose π , followed by quadrant reduction and rational Taylor intervals for sine and cosine. With $\epsilon = 2^{-24}$, every stored root component is within 4ϵ , hence every root has Euclidean error below 6ϵ . A retained word has at most 41 phase factors and

$$\frac{(1 + 6\epsilon)^{41} - 1}{\epsilon} < 247.$$

Fewer than 20,000 binary64 rounding units occur along a state path and complete projection, while $\gamma_{20000}(2^{-53}) < \epsilon$. Therefore

$$|\mathcal{Q}_{S_i, N_i}(c)u - C_i| \leq 500\epsilon s_i A_i. \quad (2)$$

The post-400 transfer remains in the same cone. With the certified untilted witness V , directed comparison gives

$$Bu \leq 3067V, \quad BV \leq 14400u, \quad H(\bar{q})V \leq 0.97V.$$

Since $s_0 + s_1 < 0.01923$ and $(0.97)^{401} < 5 \cdot 10^{-6}$, the post-400 contribution is below $1.14 \cdot 10^{-6}u$. Define

$$U = C_0 + C_1 + 500\epsilon(s_0 A_0 + s_1 A_1) + s_1 T_{25} + 8 \cdot 10^{-9} s_0 T_{41} + 1.14 \cdot 10^{-6}u. \quad (3)$$

Theorem 5.1 (Twelve-prime directed interval certificate). *For the frozen vector u ,*

$$\mathcal{R}_{<61}u \leq U < 0.933u.$$

More precisely,

$$\max_{u(x) > 0} \frac{U(x)}{u(x)} \leq 0.91717912709094451 < 0.933,$$

and $U(x) = 0$ at every allocated coordinate with $u(x) = 0$.

Proof. The Fourier centres are enclosed by (2); T_{25} and T_{41} are directed upper images; the last term of (3) covers the infinite tail. The final CUDA kernel uses upward binary64 operations for every positive sum, product and division, and compares with the downward binary64 neighbour of 0.933. It checks all 55,076,704 allocated coordinates. The threshold-failure, positive-over-zero, negative-upper, and nonfinite counts are all zero. The maximum is the displayed value. Every source, input, output, header, and length is bound by `certificate/manifest.json`; the frozen output is `certificate/admission.txt`. $\square$

6 Assembly

Theorem 6.1 (Closure of Erdős Problem #1212). *Theorem 1.1 holds.*

Proof. Lemma 3.1 and Theorem 4.2 make the all-fresh term finite in the positive necklace norm. The repeat operator is the disjoint sum of its $p < 61$ and $p \geq 61$ parts. Theorems 5.1 and 4.3 give

$$\|\mathcal{R}\| < 0.933 + 0.067 = 1.$$

Lemma 3.2 now proves $\tau(M) \rightarrow 0$ without moving the first-repeat cut. Theorem 2.3 and Proposition 1.2 yield the unbounded path in $\mathcal{S}$. $\square$

7 Certificate contract

The certificate kernel consists of one periodic-graph/Collatz checker, two directed image generators, the final directed interval checker, and the exact-rational root verifier. The two root tables are committed because they are small. The dense state images are excluded from Git because each double image is 440,613,656 bytes; their exact byte lengths and SHA-256 digests are frozen in `certificate/manifest.json`. The repository gate rejects unlisted files, missing theorem bindings, altered kernel sources, altered root tables, and any admission transcript without all four zero-failure counts. With the hash-pinned dense payload present, the full replay script rebuilds the final CUDA checker and reruns the statewise comparison.

References

- [1] P. Erdős, *A survey of problems in combinatorial number theory*, Annals of Discrete Mathematics 6 (1980), 89–115.
- [2] I. Vardi, *Deterministic percolation*, Communications in Mathematical Physics 207 (1999), 43–66.
- [3] T. F. Bloom, *Erdős Problem #1212*, <https://www.erdosproblems.com/1212>.